CHEATSHEET

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Diagnostic Scripts Cheatsheet

Quick-reference cheatsheet of diagnostic scripts and one-liners for rapid VCF component troubleshooting and health validation.

SCRIPTS | ONE-LINERS | DIAGNOSTICS | RAPID TRIAGE

VCF 9.0 VMware Cloud Foundation

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VCF 9.0.1 Diagnostic Scripts Cheat Sheet

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Overview

20 Python diagnostic scripts for VCF 9.0.1 nested lab troubleshooting. All use **Paramiko** for SSH and run from a Windows workstation.

```
pip install paramiko
python <script-name>.py
```

Connection Targets

Target	IP	User	Purpose
SDDC Manager	192.168.1.241	vcf	API gateway, database access (su to root)
NSX Node	192.168.1.71	root	Direct NSX service management
NSX VIP	192.168.1.70	admin	NSX cluster API (via curl from SDDC Mgr)

> ***Note:*** *All scripts use the standard lab password. If* `vcf` *is locked out:* `faillock --user vcf --reset` *from SDDC Manager console as root.*

Quick Reference Card

Scenario	Script
Is everything healthy?	`python quick_status.py`
NSX slow after boot?	`python nsx_monitor.py`
Credential operation failed?	`python check_remediate_error.py`
Need to update NSX password?	`python nsx_cred_update.py`
NSX CPU overloaded?	`python nsx_slim.py`
Put NSX services back?	`python nsx_restart_all.py`
Clear stale DB locks?	`python clear_locks.py`
Fix stuck tasks in DB?	`python fix_stuck_tasks.py`
Full cascade fix?	`python full_remediate_fix.py`
System clean after fix?	`python final_check.py`

Status & Health Checks (Read-Only)

Script	Connects To	What It Does
`quick_status.py`	SDDC Manager	**Start here.** NSX status, VIP health, resource locks, notifications, credentials
`final_check.py`	SDDC Manager	Lightweight: notifications and resource locks only
`diag.py`	localhost	DNS resolution, TCP 443 connectivity, ARP/routing from Windows host
`nsx_monitor.py`	NSX Node	Polls cluster status + load avg every 60s for 10 iterations

NSX Diagnostics (Read-Only)

Script	Connects To	What It Does
`nsx_check.py`	SDDC Manager	Tests both NSX VIP and direct node connectivity — diagnoses VIP failover issues
`nsx_diag.py`	NSX Node	Top CPU consumers, disk space, service health via API, catalina errors

`nsx_resource_check.py`	SDDC Manager	NSX clusters, credentials, warnings, DB resource state
`sddc_nsx_status.py`	SDDC Manager	Compares SDDC Manager's NSX status vs actual NSX VIP cluster status

Diagnostic Escalation Path

```
python quick_status.py          # 1. Overall health
python nsx_check.py             # 2. VIP + node connectivity
python nsx_diag.py              # 3. Performance & services
python sddc_nsx_status.py       # 4. SDDC Manager vs NSX sync
```

Credential Operations

Script	Modifies	What It Does
`nsx_cred_update.py`	**Yes**	Full workflow: health checks, lists credentials, updates admin API, monitors 200s
`nsx_retry_when_ready.py`	**Yes**	Waits up to 15 min for NSX API, then submits update with 450s monitoring
`check_disconnected.py`	No	Inspects all credential objects for connection status fields
`check_remediate_error.py`	No	Failed task details with full error messages, NSX connectivity test, log search

> **WARNING:** Do not run credential update scripts if NSX status is not `ACTIVE` in SDDC Manager or `STABLE` at the VIP. A failed update creates stale locks and stuck tasks, requiring database repair.

NSX Service Management

Script	Action	What It Does
`nsx_slim.py`	**Stops**	Stops 5 non-essential services to free CPU during boot storm
`nsx_restart_all.py`	**Starts**	Restarts all services stopped by nsx_slim.py
`nsx_fix_svc.py`	**Restarts**	Restarts search, nsx-sha, nsx-appl-proxy, validates health

Service Recovery Sequence

```
python nsx_slim.py              # Free CPU (if load > 30)
# Wait for load to drop below 15
python nsx_restart_all.py       # Bring services back
python nsx_check.py             # Verify cluster health
```

Database Fixes

Scripts that modify SDDC Manager's PostgreSQL database. All use the trust auth workaround (backup pg_hba.conf, set trust, fix, restore).

Script	What It Does
`clear_locks.py`	Fixes NSX status (ACTIVATING/ERROR -> ACTIVE), clears `lock` table, restarts operationsmanager
`fix_stuck_tasks.py`	Marks stuck `task_metadata` as resolved, clears `task_lock`, fixes `execution_to_task` orphans
`full_remediate_fix.py`	**Complete cascade fix:** NSX health check + DB fix (status + locks + tasks) + service restart
`find_pg_pass.py`	Searches for PostgreSQL password in config files (read-only)
`get_task.py`	Retrieves task details by ID with subtask errors (edit task_id before running)

> *Always use `PAGER=cat` when running psql to prevent pager traps in remote sessions.*

Database Repair Quick Reference

```
-- Connect: su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform"

-- Fix NSX status (covers ACTIVATING and ERROR)
UPDATE nsxt SET status = 'ACTIVE' WHERE status != 'ACTIVE';

-- Clear stale locks
DELETE FROM lock;

-- Resolve stuck tasks
```

```
UPDATE task_metadata SET resolved = true WHERE resolved = false;
DELETE FROM task_lock;
```

Full procedure: [Troubleshooting Handbook](#) Section 10

Troubleshooting Decision Tree

```
Problem: Credential operation failed
    |
    +-- python quick_status.py
    |    |
    |    +-- NSX Status = ACTIVATING or ERROR?
    |    |    +-- python clear_locks.py (fix DB status + locks)
    |    |    +-- python fix_stuck_tasks.py (resolve stuck tasks)
    |    |    +-- OR: python full_remediate_fix.py (all-in-one)
    |    |
    |    +-- NSX VIP returning 503?
    |    |    +-- python nsx_diag.py (check load)
    |    |    +-- Load > 30? -> python nsx_slim.py (free CPU)
    |    |    +-- Wait -> python nsx_monitor.py (track recovery)
    |    |
    |    +-- All green?
    |         +-- python nsx_cred_update.py (retry update)
```

Document Information

Field	Value
Document Title	VCF 9.0.1 Diagnostic Scripts Cheat Sheet
Version	2.0
Last Updated	February 2026
Script Count	20 scripts
Prerequisite	Python 3.x + Paramiko (`pip install paramiko`)
Environment	Dell Precision 7920, VMware Workstation Nested Lab

Scripts are for lab/educational purposes. Verify target hosts before running. Database-modifying scripts back up and restore pg_hba.conf automatically.

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